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BATCH 45

ASS 2.5

Task 1: Refactoring Odd/Even Logic (List Version)

❖ Scenario:

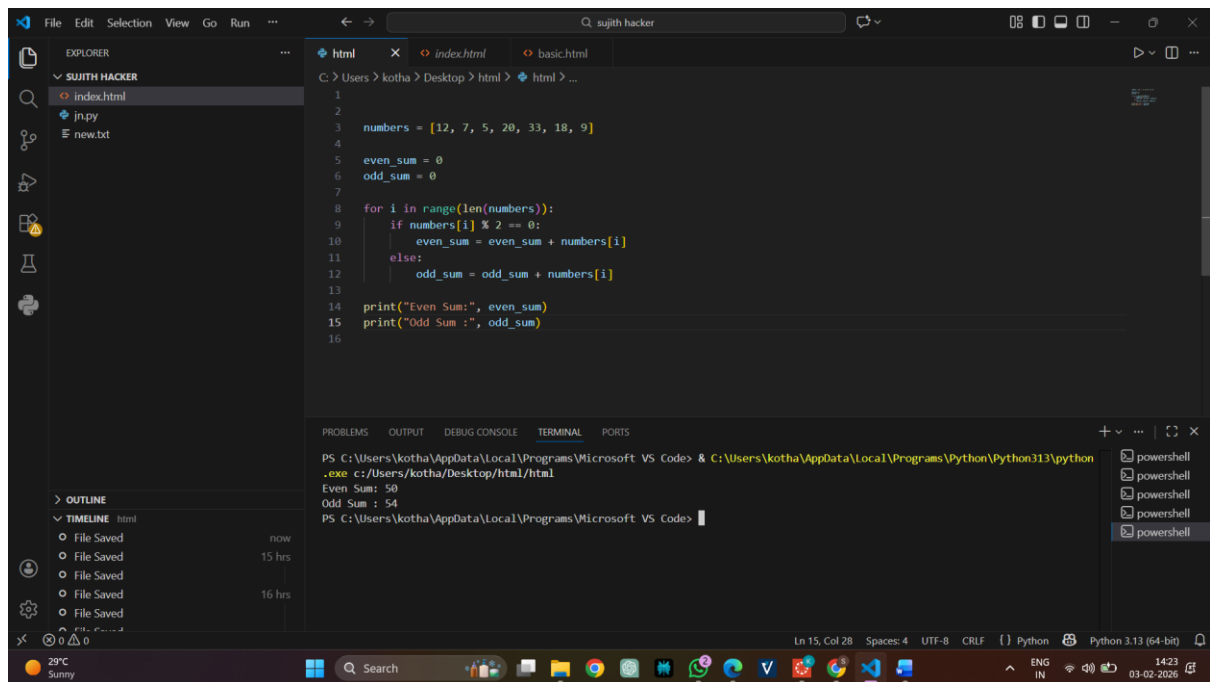
You are improving legacy code.

❖ Task:

Write a program to calculate the sum of odd and even numbers in a list, then refactor it using AI.

❖ Expected Output:

❖ Original and improved code



```
1
2
3 numbers = [12, 7, 5, 20, 33, 18, 9]
4
5 even_sum = 0
6 odd_sum = 0
7
8 for i in range(len(numbers)):
9     if numbers[i] % 2 == 0:
10         even_sum = even_sum + numbers[i]
11     else:
12         odd_sum = odd_sum + numbers[i]
13
14 print("Even Sum:", even_sum)
15 print("Odd Sum :", odd_sum)
16
```

```
PS C:\Users\kotha\AppData\Local\Programs\Microsoft VS Code> & C:\Users\kotha\AppData\Local\Programs\Python\Python313\python.exe c:/Users/kotha/Desktop/html/html.py
Even Sum: 50
Odd Sum : 54
PS C:\Users\kotha\AppData\Local\Programs\Microsoft VS Code>
```

Task 2: Area Calculation Explanation

❖ Scenario:

You are onboarding a junior developer.

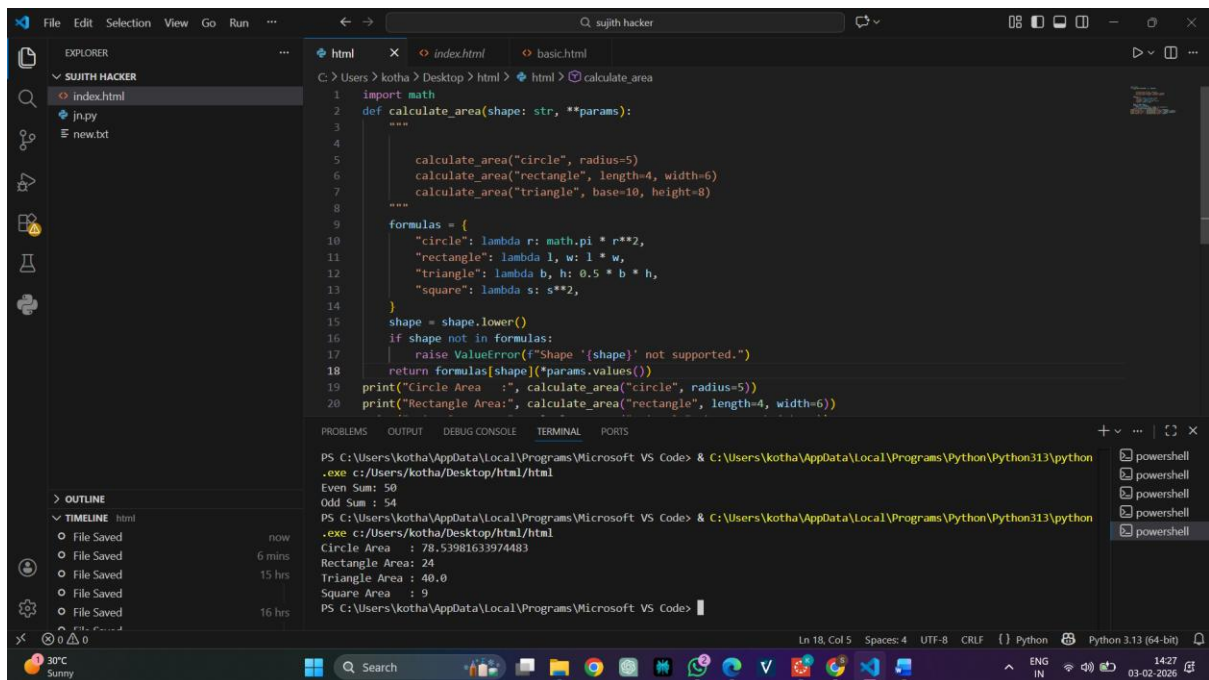
❖ Task:

Ask Gemini to explain a function that calculates the area of different shapes.

❖ Expected Output:

➤ Code

➤ Explanation



```
File Edit Selection View Go Run ...
Q sujith hacker

EXPLORER
SUJITH HACKER
  index.html
  jupyter
  new.txt

index.html
1 import math
2 def calculate_area(shape: str, **params):
3     """
4     calculate_area("circle", radius=5)
5     calculate_area("rectangle", length=4, width=6)
6     calculate_area("triangle", base=10, height=8)
7     """
8     formulas = {
9         "circle": lambda r: math.pi * r**2,
10        "rectangle": lambda l, w: l * w,
11        "triangle": lambda b, h: 0.5 * b * h,
12        "square": lambda s: s**2,
13    }
14    shape = shape.lower()
15    if shape not in formulas:
16        raise ValueError(f"Shape '{shape}' not supported.")
17    return formulas[shape](**params.values())
18 print("Circle Area :", calculate_area("circle", radius=5))
19 print("Rectangle Area:", calculate_area("rectangle", length=4, width=6))
20

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\kotha\AppData\Local\Programs\Microsoft VS Code> & C:\Users\kotha\AppData\Local\Programs\Python\Python313\python
.exe c:/Users/kotha/Desktop/html/html
Even Sum: 50
Odd Sum: 54
PS C:\Users\kotha\AppData\Local\Programs\Microsoft VS Code> & C:\Users\kotha\AppData\Local\Programs\Python\Python313\python
.exe c:/Users/kotha/Desktop/html/html
Circle Area : 78.53981633974483
Rectangle Area: 24
Triangle Area : 40.0
Square Area : 9
PS C:\Users\kotha\AppData\Local\Programs\Microsoft VS Code>

Ln 18, Col 5 Spaces: 4 UTF-8 CRLF Python Python 3.13 (64-bit)
```

Task 3: Prompt Sensitivity Experiment

❖ Scenario:

You are testing how AI responds to different prompts.

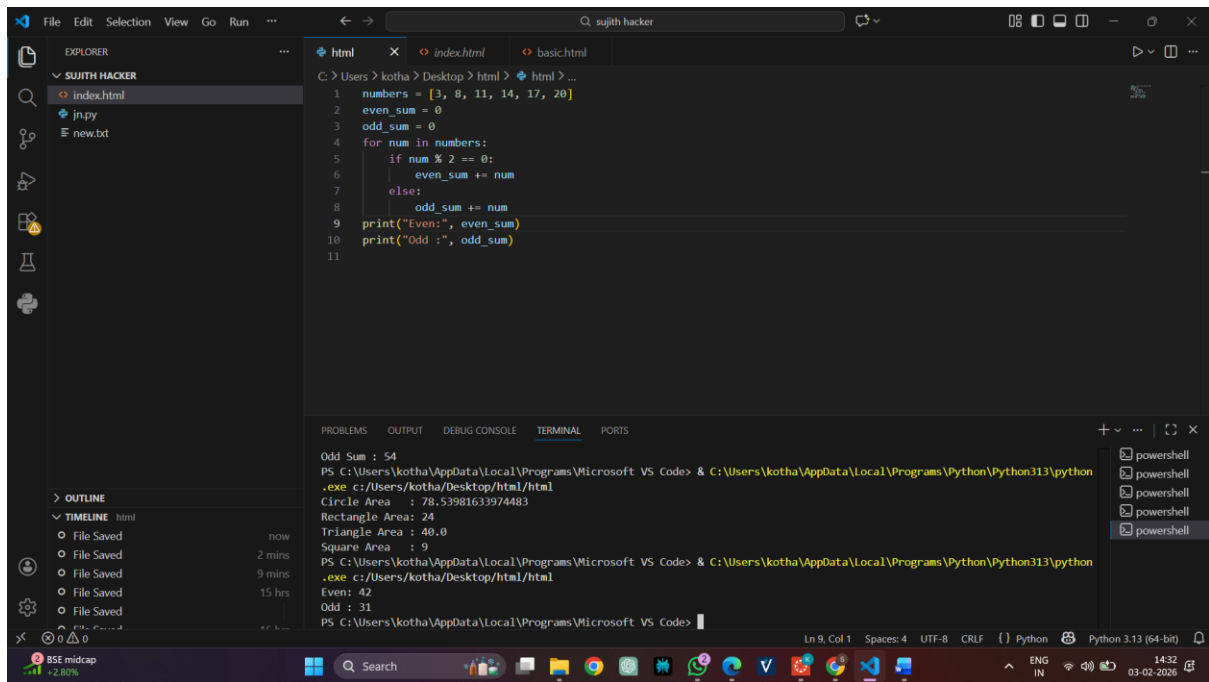
❖ Task:

Use Cursor AI with different prompts for the same problem and observe code changes.

❖ Expected Output:

➤ Prompt list

➤ Code variations



Task 4: Tool Comparison Reflection

❖ Scenario:

You must recommend an AI coding tool.

❖ Task:

Based on your work in this topic, compare Gemini, Copilot, and Cursor AI for usability and code quality.

❖ Expected Output:

Short written reflection

